

COMPETITION ENTRY · INTERNATIONAL TRACK

CarbonPass

When the AI era builds a wall of data around the world's markets, digital inclusion means handing every small factory a door.

A conversational, local-first AI that turns Taiwan's open data into verified market access for 2,600 exporting SMEs — starting in the Kingdom of Screws.

2026 Presidential Hackathon — International Track

Theme: "Digital Inclusion in the AI Era"

Corresponding SDGs: 8 · 9 · 12 · 13

Team: [3-10 members, incl. international members — to be completed]

Submission window: 22 May – 31 July 2026

Every figure in this proposal is cited to a primary or dated credible source.
Estimates are labeled. Pending legislation is presented as pending.

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1 Executive Summary

On 1 January 2026, the European Union began charging for the carbon embedded in imported steel goods. Taiwan shipped **3.74 million tonnes** of covered goods to the EU in the preceding reporting period — mostly screws and fasteners made by **about 2,600 small, family-run factories** concentrated in Kaohsiung's Gangshan district, the "Kingdom of Screws"¹. From this year, each of those factories faces a choice made in Brussels: deliver verified, product-level carbon data to its European buyer, or watch that buyer pay punitive "default values" — marked up 10% now, 30% by 2028⁸ — until the order quietly moves to a supplier who can deliver the data.

The knowledge needed to escape that penalty exists. It is buried in a 2,400-page implementing regulation⁸ that a sixty-year-old factory owner, whose back office is a family member with a spreadsheet, cannot read — and no tool in Taiwan's current ecosystem reads it for him^{1,18,20}. **This is the digital divide of the AI era:** not the absence of broadband, but the inability to produce the data that now decides who may participate in the world's markets.

CarbonPass closes that divide with a photograph. Inside LINE, in Mandarin or Taiwanese, an owner photographs an electricity bill and a folder of invoices. A local-first AI — raw data never leaves the factory — returns three things: (1) the verifier-ready, per-product carbon data pack his EU buyer must have, in the European Commission's own format; (2) one decision screen showing what his data is worth: "*with your numbers, your buyer pays ≈EUR 150 per tonne; without them, EUR 450–750 and rising*"^{17,23}; and (3) a production schedule, computed from Taiwan's open 10-minute grid data²⁶, that shifts furnaces and compressors to cheaper, cleaner hours — cutting the power bill now and the next data pack's numbers with it.

The proof of concept runs entirely on Taiwan's open data — MOENV emission coefficients, Taipower grid telemetry and tariffs, EU published methodology — and gives data back: a cleaned hourly grid-carbon-intensity dataset and an anonymized sector benchmark, published openly. A pilot factory in Gangshan is targeted for the mentorship period; measurable before/after results for the final review. If the European Parliament's September vote extends the border mechanism to ~180 further product categories¹⁰, the same engine scales with it — and, beyond Taiwan, to every SME exporting economy facing the same wall. Taiwan can help.

2 The Campaign and Its Moment

2.1 Theme, scope, and nature of the Presidential Hackathon

The Handbook defines the 2026 theme, "Digital Inclusion in the AI Era," through two interlocking objectives: *promoting digital inclusion* — "enabling people of different ages, regions, and needs to access technology and participate in digital services on equal footing, thereby narrowing the digital divide" — and *building smarter services* across "education, care services, governance, disaster prevention, and beyond"²⁹. The campaign's nature is equally explicit: initiated by the Office of the President in 2019 and echoing President Lai's vision of "advancing toward a smart and prosperous Taiwan," it seeks data-driven solutions to public issues, built on open data, carried by international and interdisciplinary teams, and intended for real adoption — selected teams receive mentorship, public-private matchmaking and field-validation support, and advancing teams join follow-up benefit tracking^{28,29}. This is not a coding contest. It is the front door of a policy-adoption pipeline — past winning work has matured into government practice³².

2.2 What history says a winning entry looks like

Reading seven editions of Teams of Excellence reveals a consistent pattern: winners are *technological answers to the most acute, well-documented anxiety of their year*, delivered on open data, with a working demonstration and a human face (Table 1).

Table 1: The winning pattern across editions of the International Track

Years	The year's anxiety	Representative winners	What made them effective
2019-20	Corruption, opaque procurement, pandemic supply chains	Cartelogy; CoST Honduras; Fiscal Force	Open contracting data turned into accountability tools
2021-22	Climate urgency, binding net-zero pledges	A/B Street; Office Farmers; BRSDM	Open data turned into carbon and adaptation action
2023-25	AI explosion; resilience of the vulnerable	HysonTech; Beyond Hearing; CropNow	Advanced AI aimed at a specific excluded population, demoted live

Three lessons carry into 2026. First, **contextual precision beats technical generality**: Beyond Hearing won not because AR was novel but because it aimed AI at a nameable excluded group. Second, **open data is the price of admission**: the application form itself asks what open data is used, what should be improved, and what will be given back²⁹. Third, **feasibility is weighted above brilliance**: 40% of the preliminary score, with a demo expected by finals²⁹.

2.3 The national moment this entry serves

Taiwan in 2026 is executing a dual transformation — digital and net-zero at once. Net-zero 2050 is statutory; the new NDC 3.0 commits to a 38%±2% cut by 2035²¹; the first carbon-fee cycle just collected NT\$4.97 billion from the island's largest emitters². At the same moment, the EU's carbon border regime began taxing the products of Taiwan's smallest exporters — and the Ministry of Environment itself has named the affected group and opened a help platform for them¹. The state has built the awareness layer; what is missing is the *computation* layer that turns awareness into survival. A Presidential Hackathon entry that completes the government's own stack, on the government's own open data, for a government-named vulnerable population, in the year the government's theme is digital inclusion — that is the alignment this proposal is built on.

3 The Problem and Its Importance

3.1 A person, not an abstraction

Consider a real profile from the Gangshan cluster: a second-generation owner in his sixties, thirty employees, precision screws for European automotive distributors, administration handled by his daughter-in-law on a spreadsheet. Nationally, 51.79% of Taiwan's SMEs are exactly this — sole proprietorships or family firms, most over eight years old¹¹. His factory is one of ~1,800 in a district that ships roughly 1.2 million tonnes of fasteners a year and holds 10-13% of world production — a top-three global exporter^{12,16}, a quarter of it bound for Europe¹³.

In January 2026 his largest buyer's procurement system began scoring him on a number he

has never produced: the verified carbon content of his screws. Sector surveys show how unprepared his cohort is: **60%** of fast-growing SMEs have completed no carbon inventory, **80%** have no reduction targets, and half name talent shortage as the binding constraint¹³.

3.2 The mechanism that turns ignorance into lost orders

The EU's Carbon Border Adjustment Mechanism entered its paying phase on 1 January 2026: importers of steel goods (including screws and bolts, CN 7318) must surrender certificates priced at the official EU rate — EUR 75.36 in Q1, EUR 75.28 in Q2 2026^{6,15,23}. With verified data, a tonne of fasteners (~2 tCO₂e embedded¹⁷) costs the buyer about **EUR 150**. Without it, punitive default values apply — 3-5× higher by industry estimate, marked up a further 10% this year and 30% by 2028^{8,17} (Figure 1). Analysts put worst-case exposure at **30-50% of product value** on commodity lines¹⁷. The 50-tonne exemption does not save the small: it is counted per *EU importer*, cumulatively across all suppliers — MOENV's Climate Change Administration has warned publicly that even minimal-volume Taiwanese SMEs will be swept in¹.

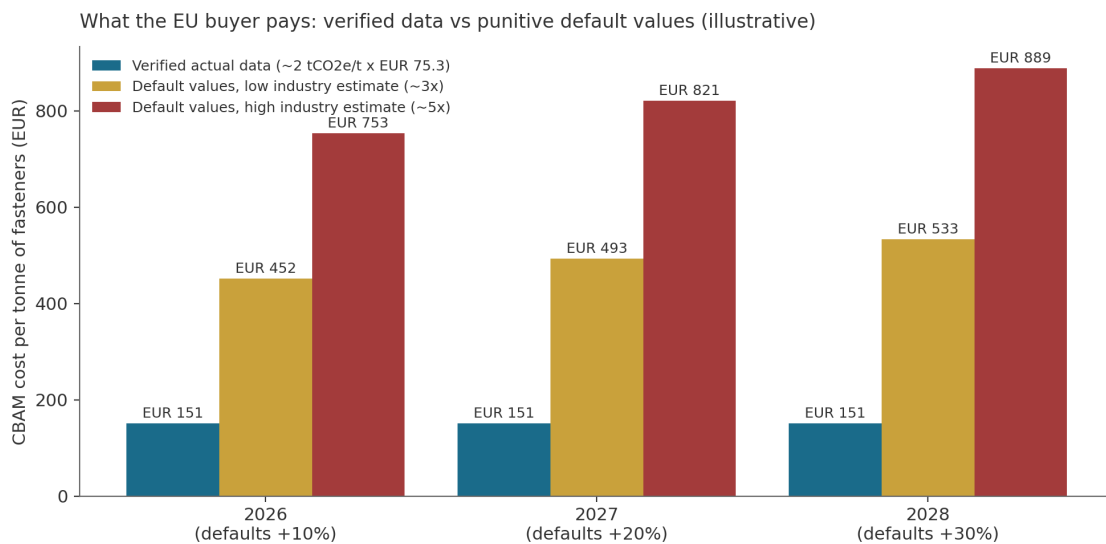


Figure 1: What the EU buyer pays per tonne of Taiwanese fasteners: verified actual data versus punitive defaults, at the official certificate price²³, with the regulation's automatic markup escalation⁸. The gap is the survival margin of a family factory.

MOENV counts **~2,600 Taiwanese SMEs** producing the affected goods — 3.74 million tonnes shipped to the EU in the last reporting period¹ (Figure 2).

Behind the firms stand the jobs, suppliers, and the economic identity of an entire southern district. And the wall is still rising: the European Parliament votes in September 2026 on extending the mechanism to ~180 further steel and aluminium product categories from 2028¹⁰, while at home the carbon-fee threshold will fall and an emissions-trading pilot approaches³¹. **This problem is not a wave but a rising floor** — and the population it lands on first has already been named by the government itself.

4 Digital Inclusion, Exactly as the Handbook Defines It

The Handbook's first objective asks entries to enable "*people of different ages, regions, and needs* to access technology and participate in digital services on equal footing, thereby narrowing the digital divide"²⁹. CarbonPass answers each axis literally:

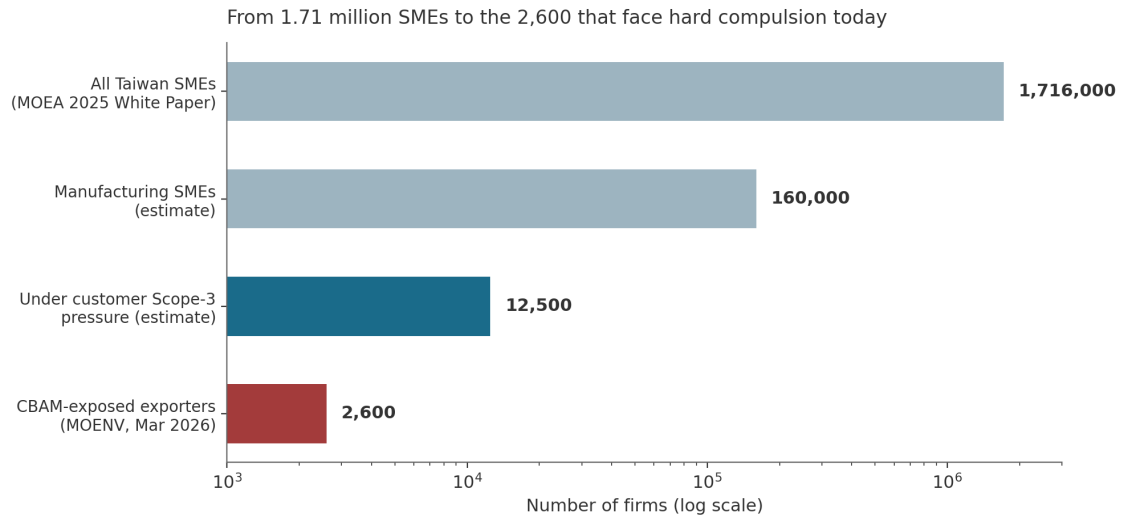


Figure 2: This proposal targets a precisely named population, not an inflated one: of Taiwan’s 1.716 million SMEs¹¹, the ones facing hard compulsion today are the ~2,600 CBAM-exposed exporters identified by MOENV¹. Middle bars are our estimates of the wider customer-pressured band.

Ages.

The population being excluded is demonstrably older and analog-native: family firms run by senior owner-operators, back offices without a single data professional^{11,13}. They are being asked to operate EU-grade data infrastructure overnight. CarbonPass meets them where they are: a voice-friendly conversation in LINE, in Mandarin and Taiwanese — “拍一張電費單” (*snap a photo of the power bill*) is the entire onboarding.

Regions.

The excluded community is geographically concrete: the Gangshan-Luzhu district of Kaohsiung, whose industrial identity — the Kingdom of Screws — is what the data wall threatens. Serving it is regional inclusion in the most literal sense.

Needs.

Carbon-verified market access has become a de facto *digital service*: participate or be de-listed. Today only corporations can afford to participate. CarbonPass extends that participation to a 12-person factory at near-zero marginal cost — equal footing, in the Handbook’s own words.

The second objective — “building smarter services ... across education, care services, governance, disaster prevention, *and beyond*”²⁹ — is answered by what the system does with public infrastructure: it turns Taiwan’s open data (emission coefficients, grid telemetry, tariffs) into a service layer that the government’s own CBAM help platform currently lacks¹.

Where earlier editions of this competition bridged the divide of broadband and the divide of disability, 2026’s sharpest divide is the data divide — and it can be bridged the same way: by putting AI on the excluded side of the wall.

5 The Solution: CarbonPass in Action

5.1 Sixty seconds in the life of the user

Mr. Lin, Gangshan, 61, opens LINE. The assistant greets him in Taiwanese and asks for a photo of last month's Taipower bill. He photographs it, then the steel wire-rod invoices, and answers three plain questions about last month's production. Ninety seconds later he has: a **carbon data pack** for his screws, per customs code, formatted to the European Commission's template and flagged line-by-line for the verifier; a **one-screen argument** he can forward to his Rotterdam buyer — *your cost with my data: $\approx$ EUR 150/t; without it: EUR 450–750/t and rising*; and a **suggestion**: shift the heat-treatment furnace to the 02:00–06:00 window this month, saving an estimated NT\$ amount on the bill and a stated number of kilograms of CO₂ — both of which flow into next quarter's pack automatically. No portal. No consultant. No data leaving his building.

5.2 The three AI systems behind the conversation

Table 2: The CarbonPass pipeline: from shoebox documents to market access

Stage	What happens
Inputs the SME already has	Taipower bills, material invoices, production records — photographed in LINE, in Mandarin/Taiwanese
Document intelligence + allocation	Vision-language parsing of messy, mixed-format documents; facility energy allocated across product lines and CN codes with quantified uncertainty
Compliance rules engine	The EU methodology (simple/complex goods, precursors, default tables from the 2,400-page regulation ⁸) encoded into field-level guidance and the Commission's template structure
Grid-aware scheduler	Hourly grid carbon intensity computed live from Taipower's open 10-minute generation feed ²⁶ + time-of-use tariffs ¹⁹ → MILP + reinforcement-learning shift plans for flexible loads
Outputs	CBAM data pack per CN code; organizational inventory; buyer-cost delta screen; monitored before/after reduction ledger

1. Document intelligence with allocation reasoning. Anyone can OCR a bill. The hard problem — the one that decides CBAM compliance — is *allocation*: splitting one facility-wide electricity number across dozens of screw variants, per customs code, defensibly. CarbonPass treats this as a constrained-inference problem over production volumes, machine ratings and operating-hour priors, and attaches a quantified uncertainty to every line, so a human verifier sees not just a number but how solid it is. This is the step no existing calculator attempts (Section 6).

2. A regulation, compiled. The EU's methodology — simple versus complex goods, precursor emissions, punitive default tables — is encoded as a rules engine that fills the Commission's own communication template and marks each field *actual* / *default* / *needs attention*. The AI does not replace the accredited verifier; it hands the verifier (and the EU importer) a structured, auditable file instead of a shoebox of paper.

3. A scheduler that reads the national grid. Taiwan publishes every generating unit's output at 10-minute resolution as open data²⁶. CarbonPass converts that feed into an hourly carbon-intensity curve for the whole grid, overlays Taipower's time-of-use prices, and optimizes when the factory's flexible loads run — subject to the owner's order deadlines, stated in chat. Savings are measured on the meter, logged in a before/after ledger structured to the baseline-and-monitoring logic of ISO 14064-2, and feed the next data pack. **Measurement and mitigation reinforce each other — the tool does not just report carbon, it removes some.**

Privacy as architecture, not promise. Taiwanese SMEs rightly treat energy and production data as trade secrets. CarbonPass is *local-first*: parsing and computation run on the factory’s own device or premises; only the outputs the owner chooses to send ever leave. The roadmap — federated benchmark learning across the cluster, hardware-backed confidential computing — follows privacy-preserving patterns Taiwan’s own financial sector has already normalized, and aligns with the national Trusted-AI posture.

5.3 Stakeholders and their roles

Table 3: Stakeholders, roles, and what each gains

Stakeholder	Role and benefit
Fastener SME owners (target audience)	Primary users; retain EU customers, cut power costs, gain a standing carbon capability
EU importers and their declarants	Receive template-ready supplier data; avoid punitive defaults ⁸
TIFI (~700 member firms) and MIRDC	Pilot recruitment, domain validation, cluster-scale rollout channel ¹⁶
MOENV / Climate Change Administration	Their CBAM help platform ¹ gains the computational layer it lacks; carbon-fee readiness grows in the SME base
MOEA / SMESA	CAAS education stack ¹⁸ gains an execution tool; supports SME competitiveness programs
Taipower	Its open data and AMI infrastructure ^{19,26} become directly value-generating for small industry
Accredited verifiers	Receive structured, uncertainty-annotated files instead of raw paper — faster, cheaper verification
Banks (green finance)	Verified SME carbon data for sustainability-linked lending

6 Differences From Existing Solutions

Taiwan’s current ecosystem is an *awareness* stack; CarbonPass is the missing *execution* layer. The audit below was verified instrument by instrument in July 2026.

CarbonPass’s difference in one sentence: it is the only tool that converts SME-native documents into importer-ready, verifier-structured, per-CN-code emissions data — repeatedly, in the owner’s own language, at near-zero marginal cost — and it is designed for *integration* with the government’s existing platforms rather than competition with them.

7 Expected Benefits (Quantified)

Corresponding SDGs: SDG 8 (protecting decent work in an export cluster), SDG 9 (industry, innovation and infrastructure), SDG 12 (responsible production), SDG 13 (climate action).

8 Open Data: Used, Improved, and Given Back

Open data is this competition’s founding ethos, and the application form asks three questions of every entry²⁹. Our answers:

Table 4: Capability audit: what exists, and what none of it does

Instrument	What it does	What it does not do
CAAS “SME Carbon Reduction Service Station” (MOEA) ¹⁸	Courses, articles, referrals	No computation; not a reporting tool
IDB Carbon Emissions Calculator (MOEA) ²⁰	Rough annual company-level CO ₂ e from bills	No product-level allocation, no CN codes, no CBAM format
MOENV CBAM Help Platform (Mar 2026) ¹	Information and consultation	Explicitly informational; computes nothing
Taipower app / eBill ¹⁹	Consumption charts, tariff simulation	No emissions conversion, no compliance artifacts
tre100 pledge platform (CIER) ²⁵	Renewable-energy commitments registry	No measurement tooling
Consultants / ESG software ³⁰	Real one-off ISO certificates	Per-engagement pricing; cannot serve recurring per-product demands across 2,600 firms

Table 5: Expected benefits, with basis of estimate

Benefit	Quantification	Basis
CBAM cost avoided per tonne of product	EUR ≈300-600/t (verified ≈150 vs. defaults 450-750, rising with markups)	Official price ²³ ; industry multiples ¹⁷
Compliance labor per data pack	Hours instead of consultant-weeks; recurring at near-zero marginal cost	Cluster ISO precedents priced per engagement ³⁰
Energy cost reduction via load shifting	Peak/off-peak tariff spread of several× on shiftable loads (est. 5-15% of bill for furnace-heavy plants; measured in pilot)	Taipower TOU schedules ¹⁹ ; pilot to verify
Emissions reduction	Measured on-meter, logged to ledger; every avoided tonne also cuts the buyer’s certificate cost	Grid EF 0.474 kgCO ₂ e/kWh ²²
Population served at beachhead	2,600 CBAM-exposed SMEs; ~1,800-firm cluster; top-3 world export industry	MOENV ^{1,12,16}
National alignment	Supports NDC 3.0 (−38%±2% by 2035) by extending measurement into the SME layer	²¹

Used. MOENV carbon-footprint emission coefficients ([data.gov.tw #28176](https://data.gov.tw/dataset/28176), OpenAPI); Taiwan product carbon-footprint registry ([#8992](#)); Taipower generation by unit at 10-minute resolution ([#8931](#), [#37331](#)); Taipower tariff schedules¹⁹; the MOEA electricity emission factor²²; EU-published CBAM methodology, default values and certificate prices^{6,23}.

Suggested improvements. (1) Publish an official hourly grid-carbon-intensity series (today it must be derived from the 10-minute unit feed); (2) publish MOENV emission coefficients mapped to customs (CN/HS) codes, so any exporter can find the factors relevant to a tariff line; (3) provide machine-readable Taipower bill formats to enable consented, automated ingestion.

Given back. CarbonPass will publish (1) a cleaned, documented hourly Taiwan grid-carbon-intensity dataset derived from [#8931](#)/[#37331](#), and (2) an anonymized fastener-sector embedded-emissions benchmark from the pilot — both openly licensed, both useful far beyond this project.

Open source. The intent is to release the compliance rules engine and the grid-intensity pipeline under an open license — the components with the greatest public-good value — consistent with the competition’s open-source encouragement²⁹.

9 Maturity, Implementation, and Verification Plan

Development stage. Early prototype, purpose-built for this competition (no prior commercialization; no modification-rule exposure). At submission: Module 1 demonstrated end-to-end on a three-firm mock corpus of real Taiwanese bill and invoice formats; the grid-aware scheduler running on a synthetic fastener-plant load profile against the *live* open 10-minute grid feed²⁶; the LINE conversational front end operational; demonstration video submitted with the application.

Validation plan — built on the campaign’s own machinery. The mentorship period is used exactly as the Handbook designs it²⁹:

Table 6: Implementation and verification milestones mapped to the official calendar

Window (Handbook)	Milestone
By 31 Jul 2026 (submission)	PoC v0 + demo video + this proposal; outreach opened to TIFI member firms, MIRDC, and Kaohsiung EPB’s voluntary-reduction advisory team as validation partners
Aug (preliminary review)	—
Mid-Sep – mid-Oct (mentorship)	One pilot factory in Gangshan: real bills in, real data pack out; pack structure reviewed with an MIRDC engineer or accredited verifier; one flexible load shifted and measured on-meter; weekly progress log
Late Oct (final review)	Live demonstration: data pack generated on stage from raw documents; live grid-aware schedule; pilot before/after ledger; updated market outlook incorporating the September European Parliament vote ¹⁰

Honest boundaries, stated up front. CarbonPass prepares verification; it does not replace the accredited verifier⁸. Preferential carbon-fee rates belong to large regulated emitters, not SMEs — the SME’s incentives are customer retention, cost pass-through, energy savings, and readiness⁴. The 2028 scope extension is pending legislation and is treated as a tracked roadmap event, not a booked benefit¹⁰. We believe this candor is itself a feature: under a rubric that weights feasibility above all²⁹, a jury should never have to discount our

numbers.

10 Future Plans: From One District to a Global Answer

2026-2027 (Taiwan, beachhead). Harden the pilot into a cluster-scale rollout with TIFI/MIRDC; propose integration of the computation layer into MOENV's CBAM help platform and SMESA's CAAS service network^{1,18}; publish the open datasets and open-source components.

2027-2028 (regulatory expansion). If the European Parliament confirms the downstream extension, onboard the newly covered categories — springs, wire, hand tools, household metal articles — from the same engine¹⁰; extend outputs toward Digital Product Passport payloads as ESPR phases in²⁷; support SMEs entering Taiwan's lowered carbon-fee threshold and ETS pilot era³¹.

2028+ (Taiwan can help). The carbon data wall is not Taiwan's alone: every SME-exporting economy — Vietnam, Thailand, Malaysia, India — faces the same mechanism with even thinner tooling. CarbonPass's architecture (open-data grid signal, local-first document AI, EU-format output) is portable by design. As past Presidential Hackathon winners carried Taiwanese civic technology to the world, this entry is built to export a Taiwanese answer to a global problem — digital inclusion as trade policy, made in Taiwan.

11 Team and Eligibility

The team will comprise 3-10 members spanning machine learning, energy systems, regulatory analysis and fastener-industry domain expertise, including at least one member of non-ROC nationality, with two designated contacts, per the Handbook's eligibility rules²⁹. All development, documentation and presentation are in English. The entry is original work, created for this competition; team intellectual-property arrangements follow the Handbook's terms, and the team commits to full participation in mentorship, presentations and follow-up benefit tracking. *[Member names, nationalities and contact details to be completed in the online application form.]*

Seven years ago this competition showed that open data could expose corruption. Three years ago it showed AI could give hearing to the deaf. In 2026, let it show that a photograph of an electricity bill, met by the right AI, can keep two thousand six hundred family factories — and the district they built — inside the world economy.

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